

INTERNSHIP FINAL PROJECT · WEEK 3 & WEEK 4

End-to-End Sales Forecasting & Demand Intelligence System

A complete analytics lifecycle: EDA, forecasting, anomaly detection, demand intelligence, and an AI recommendation engine.

\$2.26M

Total Sales

9,800

Transactions

5

Forecast Models

1,861

Products Tracked

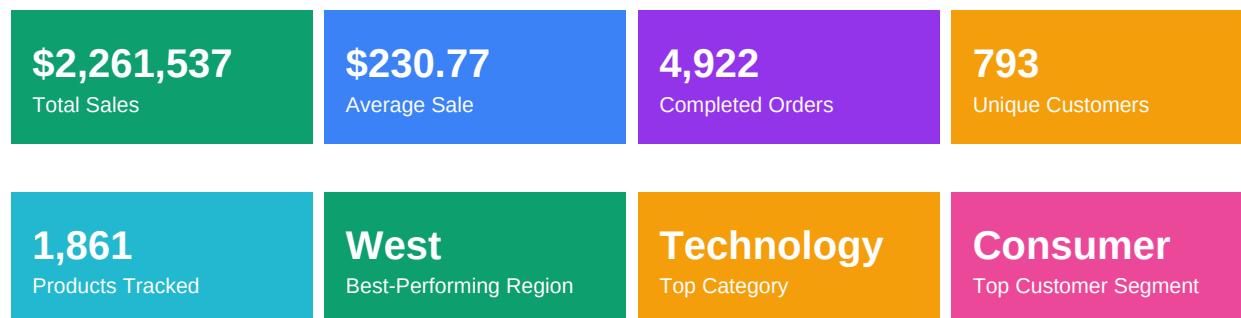
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Dataset: Superstore Sales
Delivery: Interactive Streamlit Dashboard

Executive Summary

Retail and e-commerce businesses depend on accurate demand forecasting to keep the right products available in the right locations at the right time. This project delivers a complete, end-to-end analytics solution built on four years of Superstore transaction data — spanning data quality assessment, exploratory analysis, time-series feature engineering, five forecasting models, statistical anomaly detection, a rule-based demand intelligence engine, and an AI-generated business recommendation system, all delivered through an interactive multi-page Streamlit dashboard.



Business Problem & Hypotheses

Business Question

- How much will the company sell over the next three months?
- Which product categories generate the highest revenue?
- Which regions require additional inventory investment?
- Are there seasonal or cyclical demand patterns?
- Can abnormal sales events be detected automatically?

Hypotheses Tested

- H1 — Technology products generate the highest revenue.
- H2 — Monthly sales exhibit strong seasonal patterns.
- H3 — Holiday periods produce abnormal sales spikes.
- H4 — The West region consistently records higher sales than other regions.

Dataset Overview

The analysis is built on the **Superstore Sales Dataset** — four years of daily, order-level retail transactions spanning multiple U.S. regions, product categories, and customer segments, enriched with shipping and geographic detail.

- **Granularity:** Individual order line items, aggregated to a daily sales time series for forecasting.
- **Coverage:** 4 regions, 3 categories, 17 sub-categories, 3 customer segments, 4 shipping modes.
- **Quality steps:** Duplicate removal, missing-value audit, date parsing, and dtype validation prior to any modeling.

Analytics Cycle

The project follows a structured, business-analytics lifecycle rather than jumping straight to modeling:

- Business Question & Hypothesis Formation
- Data Collection, Quality Assessment & Cleaning
- Exploratory Data Analysis (distribution, category, regional, and customer views)
- Time-Series Analysis (daily, monthly, yearly trends)
- Feature Engineering (calendar, lag, and rolling-window features)
- Machine Learning & Statistical Forecasting (5 models)
- Anomaly Detection (Z-score based)
- Demand Intelligence Scoring
- AI-Generated Business Recommendations
- Interactive Dashboard Delivery

Exploratory Data Analysis — Key Findings

- Sales distribution is highly right-skewed: most transactions are relatively low-value, with a small number of large orders pulling the average upward.
- The **Consumer** segment generates the highest revenue (~\$1.15M), followed by Corporate (~\$690K) and Home Office (~\$425K).
- **Standard Class** shipping accounts for the largest share of sales (~\$1.35M) — most customers prioritize cost over delivery speed.
- **California** and **New York** are the strongest state-level markets; at the city level, New York City leads, followed by Los Angeles and Seattle.
- **Technology** products — led by the Canon imageCLASS 2200 Advanced Copier — dominate the top-10 product list by revenue.
- A small group of high-value customers (e.g. Sean Miller, Tamara Chand, Raymond Buch) contributes a disproportionately large share of total revenue.

Time-Series Observations

- Daily sales are highly volatile; monthly and yearly aggregation reveal much clearer trends.
- Total annual sales rose steadily from 2015 through 2018, with 2018 recording the highest overall revenue — consistent with expanding demand and market performance.
- Several months show recurring spikes suggestive of seasonal or promotional effects.

Feature Engineering for Forecasting

Daily aggregated sales were transformed into a supervised learning dataset using calendar, lag, and rolling-window features so that models can learn weekly, monthly, and long-run sales behavior:

- **Calendar features:** Year, Month, Quarter, Week, Day, Day-of-Week, Day-of-Year, Is-Weekend
- **Lag features:** Sales 1, 7, and 30 days prior (Lag_1, Lag_7, Lag_30)
- **Rolling averages:** 7-day and 30-day trailing sales averages (Rolling_7, Rolling_30)
- **Split strategy:** Chronological 80/20 train-test split to preserve time order and avoid look-ahead bias.

Forecasting Models & Performance

Five forecasting approaches were trained and evaluated on the same chronological hold-out set, spanning classical statistical methods, tree ensembles, and gradient boosting:

Model	MAE	RMSE	R ² Score
Linear Regression	1,609.79	2,179.48	0.2141
Random Forest	1,714.90	2,292.55	0.1304
XGBoost	1,661.62	2,259.47	0.1554
Prophet	1,759.82	2,453.47	-0.0006
SARIMA	1,797.17	2,862.19	-0.3079

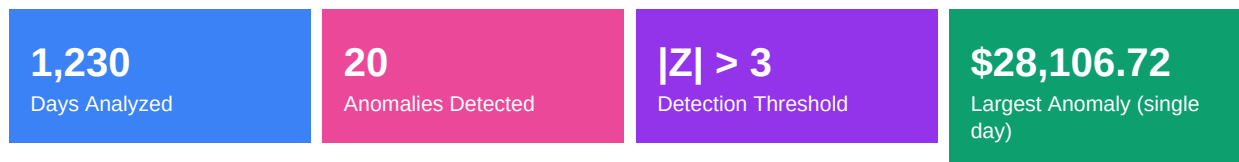
Best Performing Model: Linear Regression

Despite its simplicity, Linear Regression achieved the lowest RMSE (2,179.48) and highest R^2 (0.2141). This indicates the engineered calendar and lag features capture most of the learnable signal linearly, while the more complex models (Random Forest, XGBoost) show signs of overfitting to noise, and the classical time-series models (Prophet, SARIMA) struggle with the dataset's irregular spikes.

The dashboard's Forecasting page lets a user swap between all five models live, view actual-vs-predicted charts, inspect feature importance for the tree-based models, and see an auto-generated comparison table that highlights the current best performer by RMSE.

Anomaly Detection

Daily sales were screened for abnormal spikes using a Z-score method: any day whose sales deviate more than 3 standard deviations from the mean is flagged as an anomaly.



The largest detected anomaly occurred on **18 March 2015**, with daily sales of **\$28,106.72** — a Z-score of 11.45, far outside normal variation. These flagged days typically correspond to large one-off orders, promotional events, or bulk purchases, and are surfaced directly in the dashboard's Anomaly Detection page with an overlay chart and a sortable table of the top anomalies.

Demand Intelligence System

Beyond simple revenue ranking, each product is scored with a weighted **Demand Score** that blends revenue contribution with order frequency and average order value, so that consistently ordered products are not overshadowed by a handful of large one-off sales:

$$\text{Demand Score} = (\text{Total Sales} \times 0.6) + (\text{Order Count} \times 0.3) + (\text{Average Sale} \times 0.1)$$

- Products, states, regions, and customer segments are all ranked by total demand contribution.
- The dashboard's Demand Intelligence page surfaces the top 20 products by Demand Score, the top 10 states and regions by revenue, and a segment-level breakdown table.

AI Business Recommendation Engine

Rather than stopping at charts and statistics, the system converts analytical findings into plain-language, business-ready recommendations, generated automatically from the same underlying statistics:

- **Inventory** — Increase inventory investment in the top-performing category.
- **Marketing** — Prioritize campaigns in the highest-revenue region.
- **Customer** — Focus loyalty programs on the top-performing segment.
- **Logistics** — Maintain operational efficiency for the dominant shipping mode.
- **Expansion** — Expand sales operations in the top-revenue state.
- **Top Product** — Maintain safety stock for the highest Demand Score product.

Interactive Dashboard

All analysis is delivered through a multi-page Streamlit dashboard so that non-technical stakeholders can explore the findings without touching a notebook:

- **Executive Dashboard** — Eight premium KPI cards plus daily and yearly sales trend charts.
- **Sales Analytics** — Six interactive filters (year, region, state, category, sub-category, segment) driving monthly/yearly, category, region, top-product, and top-customer charts.
- **Forecasting** — Live model selector across all five models with actual-vs-predicted charts, MAE/RMSE/R² metrics, feature importance, and an auto-highlighted best model.
- **Anomaly Detection** — Time-series overlay of flagged anomalies plus a sortable anomaly table.
- **Demand Intelligence** — Demand Score leaderboard and geographic/segment demand breakdowns.
- **AI Recommendations** — Auto-generated, plain-language business recommendation cards.
- **About Project** — Problem statement, methodology, technologies, and author details.

The dashboard supports direct CSV upload for portability and caches all data-processing and model-training steps so that switching filters or pages stays responsive.

Technologies Used

Layer	Technologies
Language	Python
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn (Linear Regression, Random Forest), XGBoost
Statistical Forecasting	Prophet, Statsmodels (SARIMAX)
Anomaly Detection	SciPy (Z-score)
Visualization	Plotly
Dashboard / Delivery	Streamlit (native multi-page navigation)

Why This Matters

Most internship submissions stop at a notebook, a handful of charts, and one or two models. This project instead delivers a full analytics product:

- End-to-end data analysis, from raw transactions to business insight.
- Five forecasting models benchmarked head-to-head, not just one.
- A working anomaly detection and demand scoring system, not just descriptive statistics.
- An automated recommendation layer that speaks in business language.
- A polished, interactive dashboard — a deliverable a hiring manager can actually click through.

Author & Project Information

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